Giri's Tech Hub, Pune.

Core Java Machine Test

Date : 07/03/2026
Time : 08 to 09.30 Am.

Instructions:

1. Solve any 4 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a program to find the third largest element in a given integer array. If the array contains duplicate values, consider only distinct elements.
Input: n = 7 Array = 12 35 1 10 34 35 12
Output: Third Largest Element = 12

Q2.Write a program in java to find a subarray with given sum from the given array?
Output :

```
The given array is : 3 4 -7 1 3 3 1 -4
[0..1] -- { 3 4 }
[0..5] -- { 3 4 -7 1 3 3 }
[3..5] -- { 1 3 3 }
[4..6] -- { 3 3 1 }
```

Q3.Given an array nums of n integers, return *an array of all the unique quadruplets* [nums[a], nums[b], nums[c], nums[d]] such that:

- **0 <= a, b, c, d < n**
- **a, b, c, and d are distinct.**
- **nums[a] + nums[b] + nums[c] + nums[d] == target**

You may return the answer in any order.
Example 1:

Input: nums = [1,0,-1,0,-2,2], target = 0
Output: [[-2,-1,1,2],[-2,0,0,2],[-1,0,0,1]]

Q4.Write a program to find the second non-repeating character in a given string.
Input: programming
Output: Second Non-Repeating Character = o

Q5.Write a program to remove all spaces from a given string.
Input: Java Programming Language
Output: JavaProgrammingLanguage

Q6. Write a program to remove duplicate characters from a string while keeping the first occurrence.
Input: programming
Output: progamin

-----ALL THE BEST-----